

SCENARIO:48

Incident Escalation & On-Call Rotation Engine

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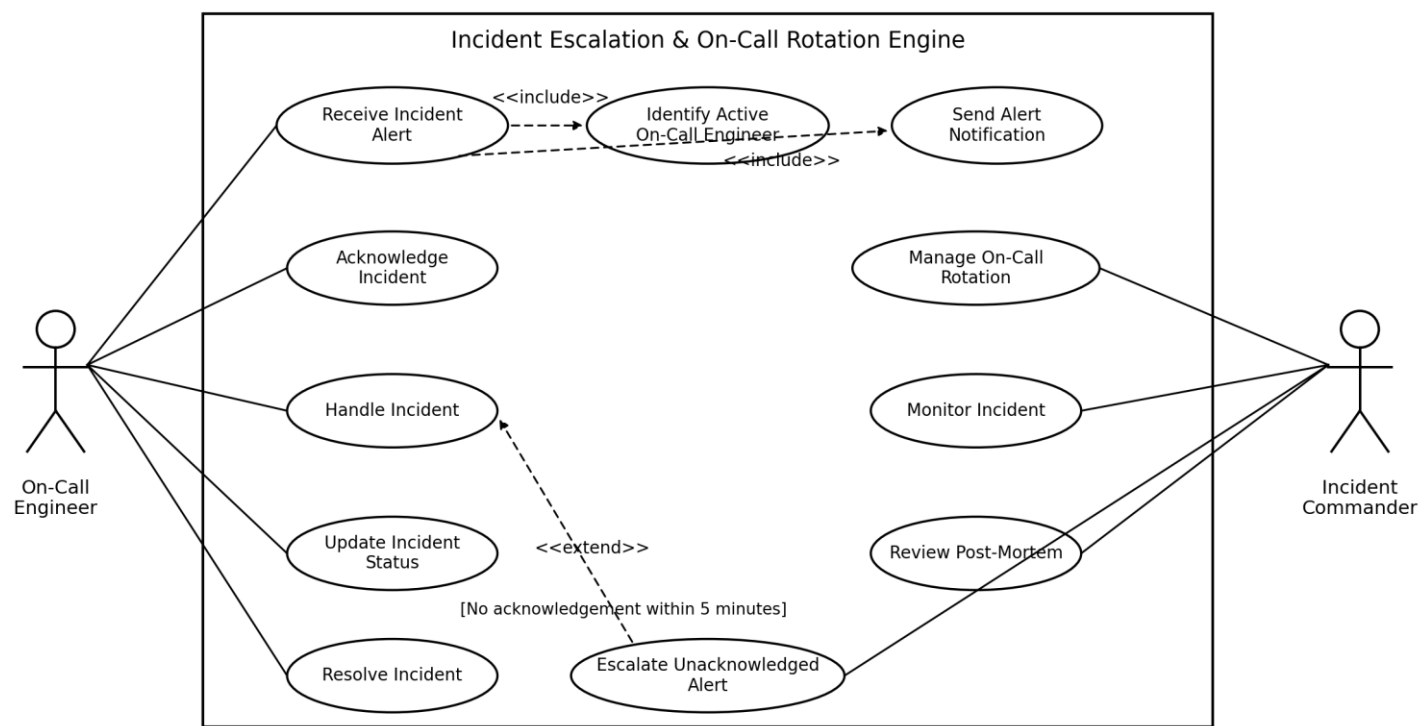
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SECTION: E

1. Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall receive incident alerts and identify the active on-call engineer from the current rotation schedule.	High	For every valid alert, the system identifies the engineer assigned to the active shift and routes the incident to that engineer.	Ensures incidents are routed to the correct person without manual checking.
FR-002	Functional	The system shall notify the active on-call engineer through configured channels such as SMS, phone, or email.	High	After an incident is received, a notification is sent to the assigned engineer through the configured communication channels.	Ensures the responsible engineer is informed quickly.
FR-003	Functional	The system shall escalate an unacknowledged incident to the secondary on-call engineer after 5 minutes.	High	If the primary engineer does not acknowledge the incident within 5 minutes, the system automatically notifies the secondary engineer.	Prevents critical incidents from being missed.
FR-004	Functional	The system shall allow the Incident Commander to create and update primary and secondary on-call rotation schedules.	Medium	Changes to the rotation schedule are saved and used when routing subsequent incident alerts.	Keeps alert routing consistent with team availability.
FR-005	Functional	The system shall store incident status, resolution details, and post-mortem information for completed incidents.	Medium	After resolution, authorized users can view the recorded incident history, resolution, and post-mortem details.	Supports review and learning from previous incidents.
NFR-001	Non-Functional — Performance	Alert dispatch through webhook, SMS, and email shall initiate within 3 seconds of alert ingestion.	High	During performance testing, alert dispatch begins within 3 seconds under normal and simulated peak conditions.	Fast dispatch is required to reduce incident response time.
NFR-002	Non-Functional — Security	The system shall allow only authenticated and authorized users to access incident data, rotation schedules, and escalation settings.	High	Users without the required permissions cannot view or modify restricted incident and rotation information.	Protects sensitive operational data and prevents unauthorized changes.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: Handle and Escalate Incident Alert

Use Case ID	UC-01
Use Case Name	Handle and Escalate Incident Alert
Primary Actor	On-Call Engineer
Supporting Actors	Incident Commander

Preconditions

1. The on-call rotation schedule is configured.
2. A primary and secondary on-call engineer are assigned.
3. The monitoring system is connected to the incident management system.
4. Notification channels such as SMS, phone, and email are configured.

Postconditions

On Success:

- The incident is assigned to the active on-call engineer.
- The incident acknowledgement is recorded.
- The incident is handled and resolved.
- The resolution details are stored.
- The incident is available for post-mortem review.

On Failure:

- The primary engineer's missed acknowledgement is recorded.
- The incident is escalated to the secondary on-call engineer.
- The incident remains active until it is acknowledged and handled.

Main Success Scenario

1. The monitoring system generates an incident alert.
2. The system receives and records the alert.
3. The system checks the current on-call rotation schedule.
4. The system identifies the active primary on-call engineer.
5. The system assigns the incident to the primary engineer.
6. The system sends the alert through the configured notification channels.
7. The On-Call Engineer receives the incident alert.
8. The On-Call Engineer acknowledges the alert within 5 minutes.
9. The system records the acknowledgement.
10. The On-Call Engineer investigates the incident.
11. The On-Call Engineer updates the incident status.
12. The On-Call Engineer performs the required corrective action.
13. The incident is resolved.
14. The system records the resolution details.
15. The incident is stored for post-mortem review.

Alternate Flow

A1 — Primary On-Call Engineer Does Not Acknowledge Alert

1. The system sends the incident alert to the primary on-call engineer.
2. The primary engineer does not acknowledge the alert within 5 minutes.
3. The system detects that the acknowledgement timeout has expired.
4. The system records the missed acknowledgement.
5. The system identifies the secondary on-call engineer from the rotation schedule.
6. The system escalates the incident to the secondary engineer.
7. The system sends the incident alert to the secondary engineer.
8. The secondary engineer acknowledges the alert.
9. The system records the acknowledgement.
10. The incident handling process continues from the investigation step in the Main Success Scenario.